1 **(a)** The fragments from the chymotrypsin digestion are more reliable than that from the trypsin digestion as the total number of amino acids listed in the chymotrypsin digestion is 46 and that it is clear that the N- and C- terminus are both Lys.

Lys-Ser-Cys-Trp-Pro-Asn-Thr-Gln-His-Arg-Asn-Ile-Tyr- Asn-Thr-Cys-Arg-Phe- Gly-Gly-Gly-Ser-Arg-Glu-Val-Cys-Ala-Ser-Leu- Ser-Gly-Cys-Lys-Ile-Ile-Ser-Ala-Ser-Thr-Cys-Pro-Ser-Tyr-Pro-Asp-Lys

[2]

(b) The primary structure may involve amino acids that have amide side chains. We might end up mistaking glutamine (Gln) for glutamic acid(Glu).

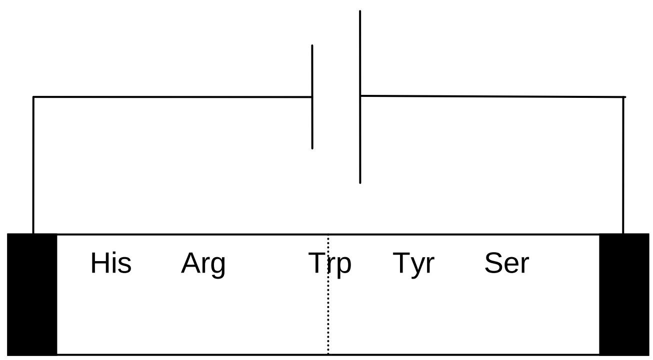
[1]

(c) **(i)** The more basic the R group the higher the pI value.

Arginine has the most basic R group as the conjugate acid exists in three resonance forms.
Tryptophan has the least basic R group as the lone pair on the N atom is delocalized into the aromatic system.
Histidine has intermediate basicity as the lone pair of N is not delocalized like in tryptophan (hence more basic than tryptophan) but cannot be stabilized by resonance (hence less basic than arginine).

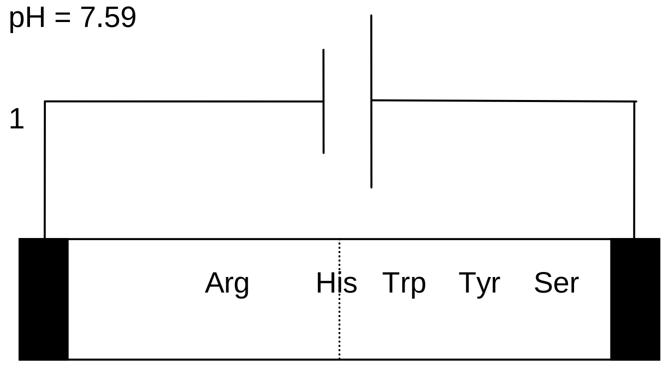
[3]

(ii) (I) pH = 5.89



starting point

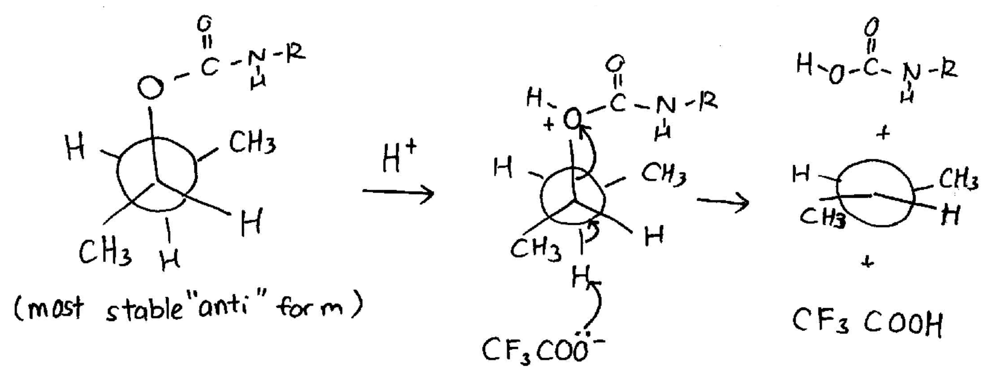
(II) pH = 7.59



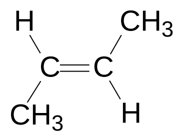
starting point

[4]

1 **(d)**

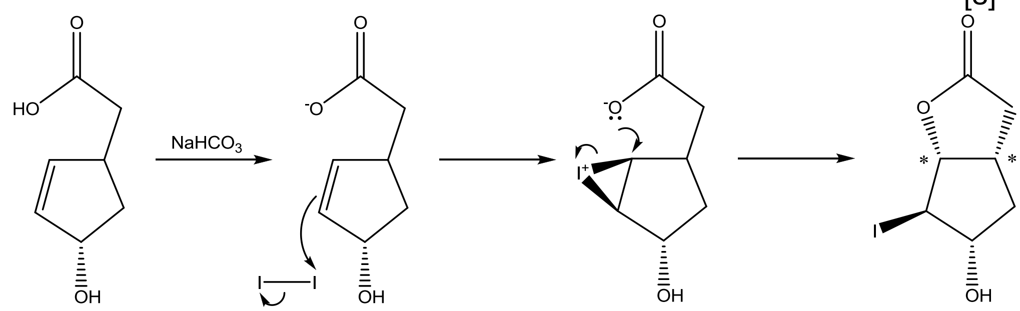


Product is trans alkene



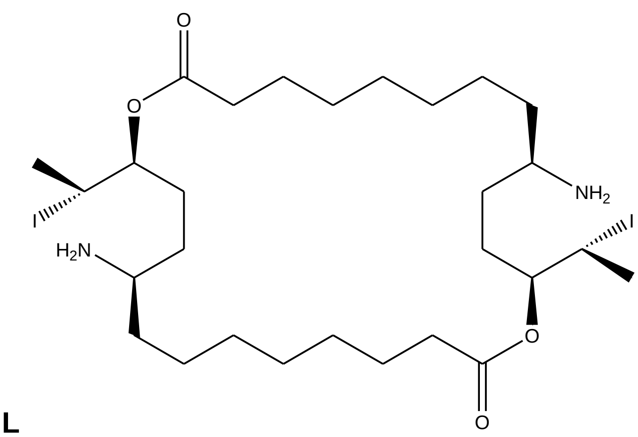
[3]

(e) **(i)**

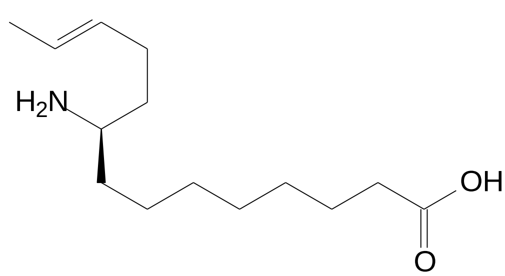


[3]

(ii)



L



K

[4]

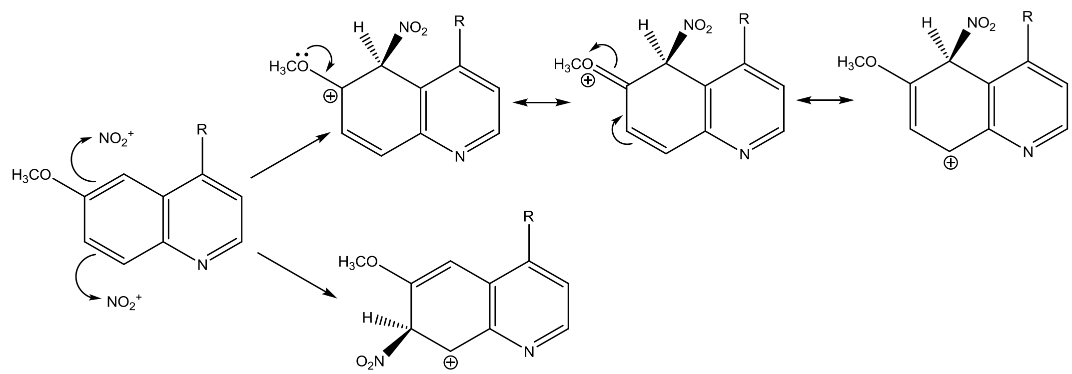
2 **(a)** **(i)** S [1]

(ii) Pyridine is less susceptible towards electrophilic attack than benzene due to decreased electron density around the ring which in turn is due to the presence of the N atom in the pyridine ring. Hence, the NO_2^+ electrophile is unlikely to attack at C1 or C2.

The –OCH_3 group on the benzene ring is 2,4 directing, favouring nitration at C5 or C7 instead of C8.

[2]

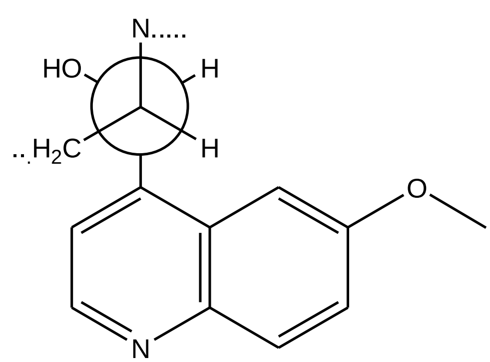
(iii) By comparing resonance structures of the two possible intermediates, the attack at C5 has 3 resonance structures which do not disturb the aromaticity of the pyridine ring. In contrast, the attack at C7 has only 1 such possible structure.



[2]

(b) **(i)** E isomer. Most stable anti-periplanar conformation of quinine is as follows.

.



[2]

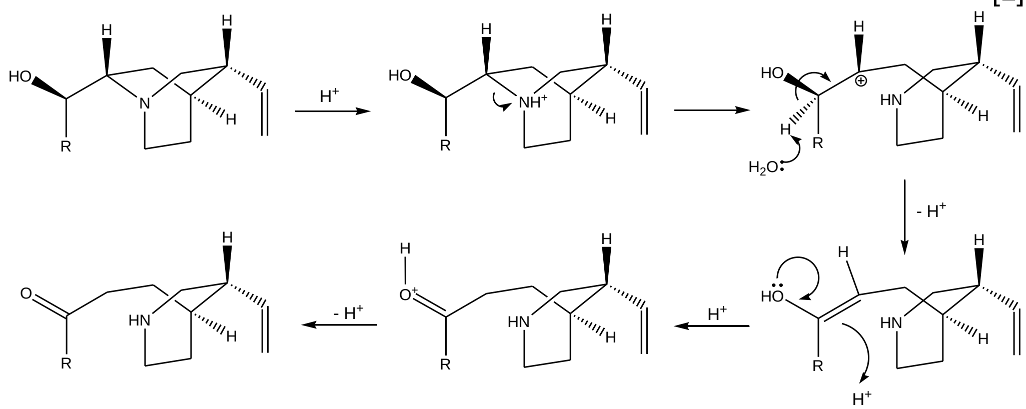
(ii) In an aqueous solvent, the E1 mechanism is favoured over the E2 mechanism. The polar solvent helps to stabilized the carbocation intermediate. The reaction kinetics become first order with respect to [quinine] and mixtures of the E and Z isomer are produced.

[2]

(c) **(i)** The N atom in quinuclidine has its three alkyl arms stretched backwards, leaving the lone pair of electrons more exposed than trimethylamine and hence a greater basicity.

[1]

(ii)



[2]

(d) **(i)** The graph A belongs to quinine and the graph B belongs to sodium benzoate.

[1]

Quinine has a more conjugated system of pi electrons as compared to sodium benzoate. It has a smaller energy gap between the HOMO and LUMO, hence it absorbs UV light at a higher wavelength. By the same token, sodium benzoate is likely to absorb at a much lower wavelength as compared to quinine.

[1]

(ii) At 340nm, using $A_{sample}/A_{initial} = c_{sample}/c_{initial}$,
$1.404/0.80 = c_{sample}/7.5 \times 10^{-5}$,
c_{sample} = **<u>1.32×10^{-4} mol dm^{-3}</u>**.

[1]

(iii) (i) Number of cans = $(2 \times 300 \times 10^{-3} / 324) / (1.32 \times 10^{-4} \times 175 \times 10^{-3})$ = **<u>80</u>**

[1]

(iv) At 240nm, absorbance due to quinine <u>only</u> is found using A = εcl,
so $A = (4700)(1.32 \times 10^{-4})(1.0) = 0.619$

Therefore, absorbance due to sodium benzoate <u>only</u> = 0.920 – 0.619 = **<u>0.301</u>**

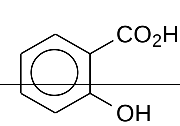
Using $A_{sample}/A_{initial} = c_{sample}/c_{initial}$,
$0.301/0.70 = c_{sample}/7.5 \times 10^{-5}$,
c_{sample} = **<u>3.23×10^{-5} mol dm^{-3}</u>**.

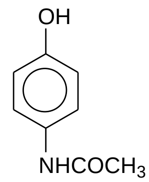
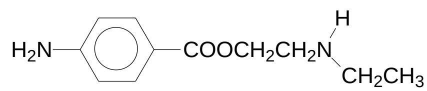
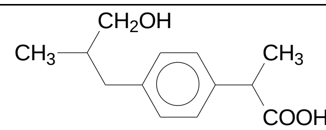
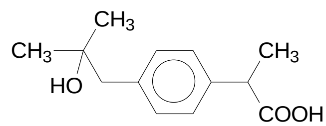
[2]

3 **(a)** **(i)** Aspirin has anti-inflammatory and anticoagulant effect but not paracetamol.

[1]

(ii)

Drug	Transformation	Metabolites
aspirin	hydrolysis	

phenacetin	O-dealkylation	
procaine	N-dealkylation	
ibuprofen	aliphatic hydroxylation	 Or 

[4]

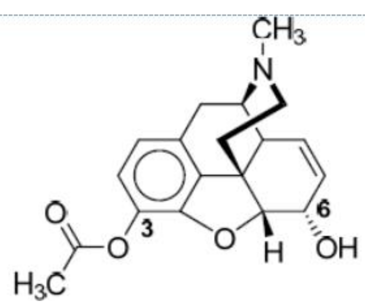
(b) **(i)** Heroin is the prodrug of morphine.

[1]

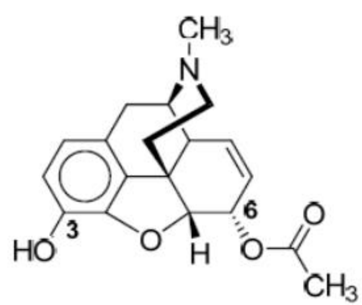
(ii) Heroin with the two hydrophilic –OH groups of morphine masked as the much less polar ester group, is less polar than morphine, is able to penetrate the hydrophobic blood-brain barrier much faster.

[1]

(iii)



3 MAM



6 MAM

(iii) The stationary phase is a layer of polar H_2O molecules. The polar molecules would have longer retention time in paper chromatography.

At pH 5.1 (acidic), the tertiary amine in heroin, morphine, 3-MAM and 6-MAM will be protonated to give a positive ammonium ion, while caffeine where the nitrogen lone pairs are either involved in aromatic delocalisation or not readily donated due to presence of numerous electronegative N atoms on the ring, remains neutral, hence least polar.

Morphine with 2 –OH group will be the most polar; 3-MAM with an alcoholic –OH will be more polar than 6-MAM with a phenolic –OH; heroin with no hydroxyl group (but esters) will be least polar.

A : 6-MAM, **B** : heroin, **C** : caffeine.

[4]

(c) **(i)** Narcotic analgesics work by targeting the opiate receptors in the brain, depressing the CNS, and hence affecting the capacity of the brain to sense pain; non-narcotic analgesics work by intercepting the pain stimulus at its source by interfering with the production of prostaglandins, chemicals that cause pain. [2]

(ii) Codeine is more soluble in water than dextromethorphan as it has additional -OH and ether group than can form more extensive hydrogen bonding with H_2O.

[2]

(d) **(i)** Student B is wrong because while precursor A is indeed a tertiary alkyl halide, it will not undergo S_N1 as the carbocation formed will not be too stable due to angle strain. Even if the carbocation can form, it will not form a racemic mixture as the direction of attack by the secondary amine is restricted by the conformation.

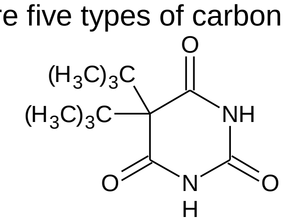
[2]

(ii) Label the **<u>axial</u>** hydrogen (i.e. replace it with D) on one of the carbons adjacent to the carbon bonded to chlorine. If it indeed were E2, the final alkene product will not contain D.
If the product obtained is a mixture of deuterated and non-deuterated alkene, it is likely to be E1

[2]

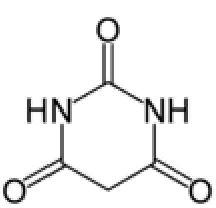
4 **(a)** They are generally achiral compounds. They are only chiral when at least one of the two R groups is/are chiral. [2]

(b) H NMR shows there is only one type of proton (highly shielded)
C NMR shows there are five types of carbon



[2]

(c)



[1]

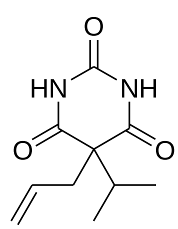
(d) **(i)** From MS spectrum, No of carbon atoms = $\frac{0.3}{2.7}\left(\frac{100}{1.1}\right) = 10$ [1]

(ii) From MS spectrum:
Peak at m/z = 167 ⇒ Loss of 210 – 167 = 43 ⇒ loss of CH_3CHCH_3
Peak at m/z = 41 ⇒ $[CH_2{=}CHCH_2]^+$ fragment

chemical shift / ppm	Integration	multiplicity	Deduction
1.0	6	Doublet	2 equivalent CH_3 group next to CH group
2.3	1	Multiplet	CH group attached to 2 equivalent CH_3
2.4	2	Doublet	CH_2 next to CH
5.0	2	Multiplet	H of $CH_2{=}C$ group
5.7	1	Multiplet	H of CH=C group
10.0	2	Singlet	H attached to N

H attached to N is highly deshielded as it is next to two electron withdrawing C=O group.

wavenumber / cm^{-1}	deduction
3200	N–H stretch
1700	C=O stretch

Aprobarbital is 

[5]

(e) (i) A ‘stimulant’ is a that acts on the nerve synapses of the central nervous system to “wake up” the central nervous system**.** This results in increases in heart rate and blood pressure. (accept also increased respiratory rate, blood sugar levels, increased adrenaline levels, disturbed sleep etc)

[1]

(ii) By reacting with an acid (*e.g.* dil HC*l*) and forming the ammonium salt (cocaine hydrochloride) that can form more favourable ion-dipole interaction with water.

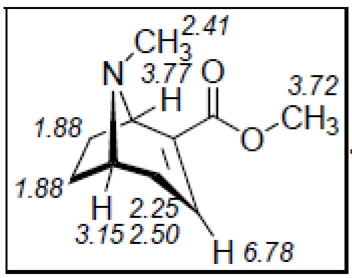
[2]

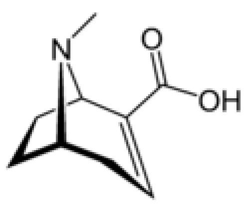
(iv) Smoking cocaine provides a higher concentration in the blood. The soluble form of cocaine being positively charged is very polar, will need to be deprotonated before it can pass through non-polar cell membranes and enter into the blood stream, while the cocaine from smoking if already in the less polar neutral form, that passes through the cell membranes more easily.

[2]

4 **(f)** M_r of cocaine = 303.
Difference between cocaine and methylecgonidine is a (303–181 = 122), $C_7H_6O_2$, benzoic acid.

δ / ppm	No. of ^{1}H	Multiplicity	No. of neighbouring ^{1}H	Structural fragement
6.78	1	Triplet	2	–CH_2–C**H**=C–
3.77	1	Triplet	2	–CH_2–C**H**–C=C–
3.72	3	Singlet	0	–OC**H**$_3$
3.15	1	Quintet	4	–CH_2–C**H**–CH_2–
2.50	1	ddd / multiplet	–	–C**H**$_2$–CH=C–
2.41	3	Singlet	0	–NC**H**$_3$
2.25	1	ddd / multiplet	–	–C**H**$_2$–CH=C–
1.88	4	Multiplet	–	–C**H**$_2$–C**H**$_2$–

Methylecgonidine is .

Ecgonidine is 

[4]

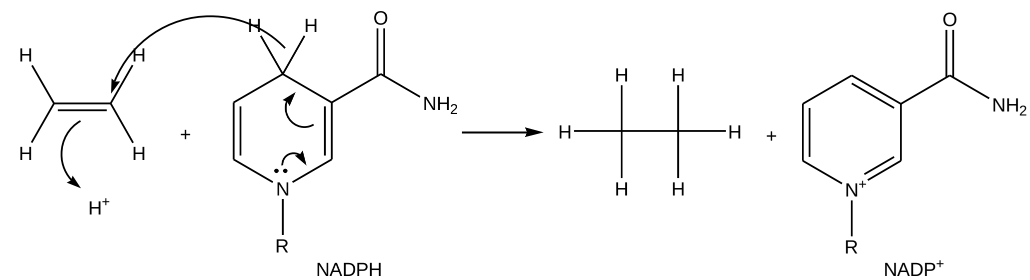
5 **(a)** **(i)** Dihydrofolate binds to the active sites of enzyme.
At pH 7.4, the –COO^- group can form ionic interaction with the –NH_3^+ at the receptor sites.
The protonated ammonium group can form ionic interaction with the –COO^- at the receptor sites.
The aromatic ring can form van der waals interaction with the non-polar sites of the receptor.
Polar -N-H and -O-H groups can form H bonding with polar groups on the receptor.

[2]

(ii) Methotrexate is structurally similar to dihydrofolate. It is a competitive inhibitor of the enzyme dihydrofolate reductase. This lowers the rate at which dihydrofolate enters the active site and gets reduced to tetrahydrofolate.

[2]

(b)



[2]

(c) Any two of the following; outline with appropriate example:

Disruption of cell metabolism
Disruption of protein synthesis
Disruption of function of plasma membrane
Disruption of nucleic acid transcription [2]

(d) **(i)** Prolong and extensive use of antibiotics allows some bacteria cells to gain the ability to combat the action of penicillins. This group of bacteria has a selective advantage of survival. The resistant strains are then able to quickly populate and evolve to be drug resistance.

(ii) Bulky ortho-methoxy groups on the aromatic ring of methicillin act as steric shields – prevents it from binding effectively to active site of penicillase (or β lactamases), thus methicillin is not hydrolysed to the inactive form, penicilloic acid, by bacterial enzyme.

[2]

(iii) Antibacterial activity of methicillin would be lower than the analogue. The analogue is expected to have greater resistance to enzymatic hydrolysis by β lactamases since the larger ethoxy group provides a larger steric shield.

Or,

Antibacterial activity of methicillin would be higher than the analogue. Level of antibacterial activity of the analogue may be lower because larger steric shield may hinder its ability to attach to the transpeptidase enzyme, thus inhibiting cell wall cross-linking to a greater extent than methicillin itself .

[2]

(iv) Penicillin G and methicillin cannot be taken orally, needs to be injected otherwise the penicillins are easily hydrolysed in the stomach/sensitive to acid-catalysed hydrolysis.

Or,

Penicillin V has electron-withdrawing O atom while ampicillin has the basic NH_2 group in the acyl side chain; both these groups reduces the nucleophilic effect of the carbonyl O atom on the acyl side chain in facilitating hydrolysis of the β lactam ring.

[1]

(f) no. of C atoms present = $\dfrac{100(A_{M+1})}{1.1(A_M)}$

$$= \frac{100(3)}{1.1(22.5)}$$

$$= 12$$

From the m/e value of the molecular ion of **B** and no. of carbon (from above) and hydrogen (from ^{1}H NMR), the remaining molar mass of **B** can be deduced to be
235 – [12(12.0) + 13(1.0)] = 78.0

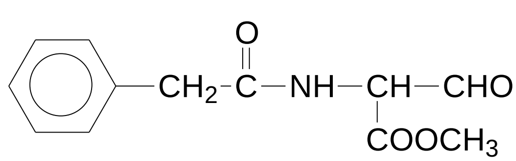
Since the molar mass of **B** gives an odd number => probably presence of 1 N atom and [78.0 – 14.0 = 64.0] hence molecular formula of **B** is $C_{12}H_{13}NO_4$.

Degree of unsaturation for **B** = 7

δ / ppm	multiplicity	integration ratio	deduction
1.7	singlet	1	-CH group with no adjacent proton
2.0	singlet	2	$-CH_2$ group with no adjacent proton
4.2	singlet	1	presence of labile proton -NH / OH group
4.5	singlet	3	$-CH_3$ group with no adjacent proton next to highly electronegative atom
7.2 – 8.0	multiplet	5	presence of monosubstituted benzene
9.1	singlet	1	presence of aldehyde

Hence using above deduction and information given about penicillin G and compound **A**,

structure for **B**:

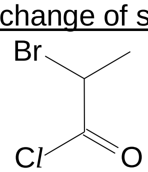


[5]

6 **(a)** An antagonist is a drug that <u>docks well at the binding site</u> but <u>does not create the required change of shape</u> of the receptor. [1]

(b)

Step 2



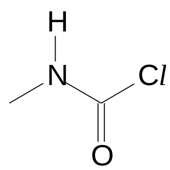
r.t.p

Step 3 Anhydrous $FeBr_3$

Step 5 H_2(g) with Ni, high temp and

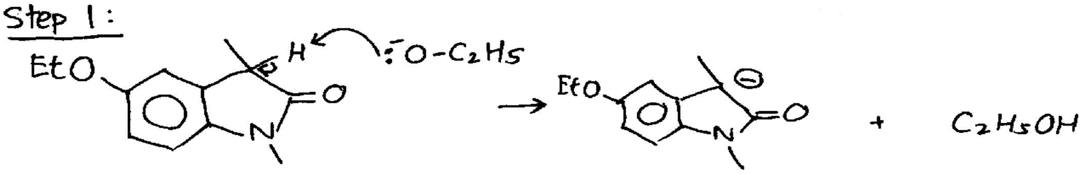
Step 8 HBr(aq) or HI(aq), heat

pressure OR H_2(g), Pt
($LiAlH_4$ will also reduce the amide group)

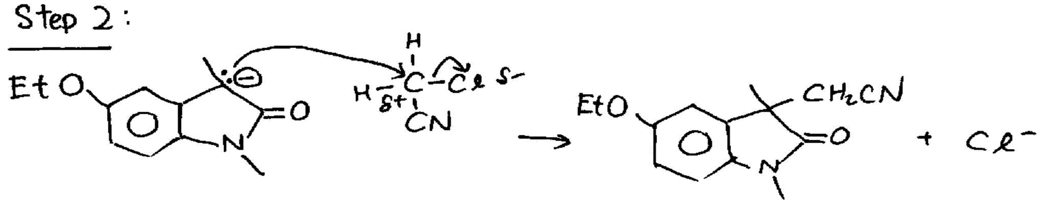
Step 9 

, r.t.p

(c) Step 1:



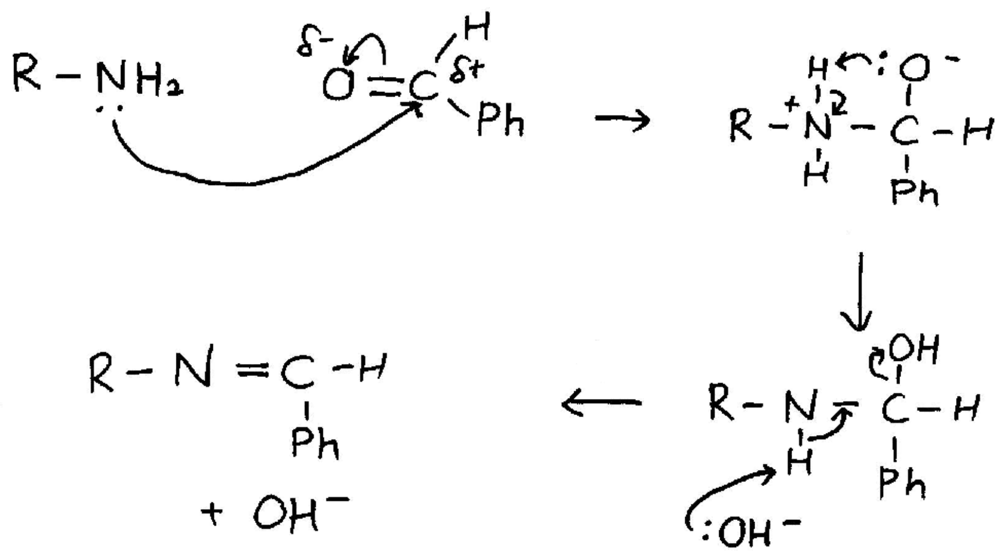
Step 2:



[2]

(d) Step 2 will fail as phenol can also undergo condensation with acyl halides.
Step 4 will fail as phenol will compete with the alpha-carbonyl carbon for the ethoxide (strong base). [2]

(e) **(i)**

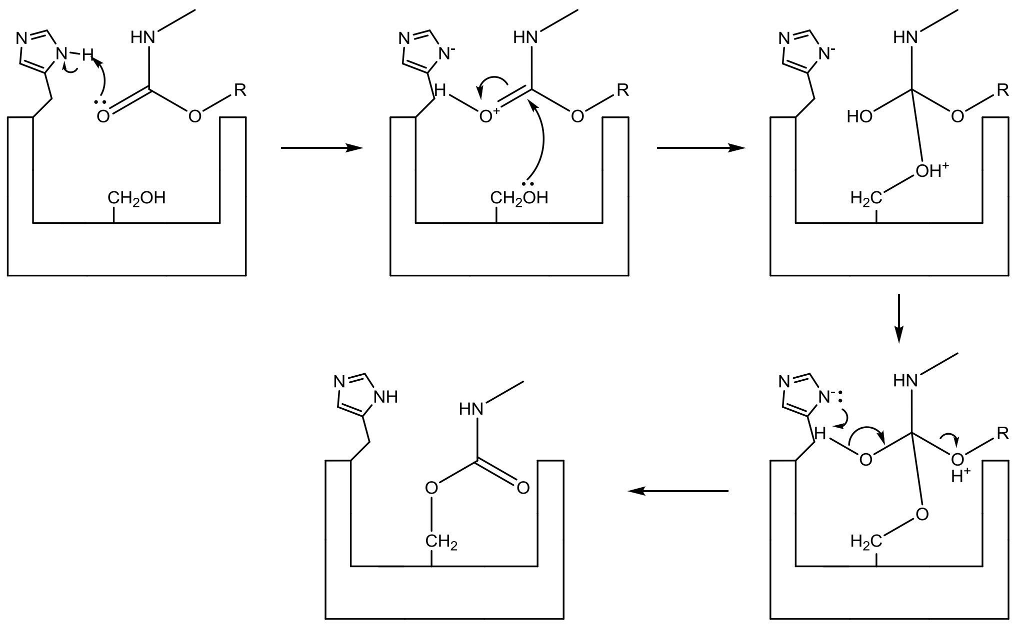


[2]

(ii) The nucleophilic substitution reaction is difficult to control and the poly-methylated product is likely to be obtained instead.

[1]

(f) **(i)**



[3]

(ii) As physostigmine is an irreversible inhibitor of acetylcholinesterase, the natural substrate acetylcholine is unable to enter the active site and be hydrolyzed. This leads to an accumulation of acetylcholine at the receptors. In the case of an atropine poisoning, the atropine competes with acetylcholine for the receptors as atropine is an antagonist (Read first sentence of paragraph 1). Physostigmine elevates the concentration of acetylcholine, causing it to displace atropine from the receptors and resume normal function.

[2]

(iii) The adjacent nitrogen group is able to donate its lone pair of electrons to the carbonyl group through delocalization of electron, this make the carbonyl carbon atom less electron deficient (less δ+) and less susceptible to hydrolysis by nucleophilic H_2O.

[2]